Windows System Hardening Project

Based on CIA Triad principles

Name: Muhammad Afaq Abbas

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1. Introduction

Windows Hardening is about making the Operating Systems more secure against potential attackers by reducing the number of possible ways to compromise or exploit them, and by minimizing the surface area of the OS that can potentially be attacked. This is achieved by configuring Security settings to appropriately protect them from potential threats to the Operating System.

This project was conducted on a Windows 11 system with the aim of improving its security posture through the implementation of practical security controls based upon the CIA Triad principles:

- **Confidentiality** - this is to protect sensitive information from unauthorized access.
- **Integrity** - this is to stop unauthorized change to data and/or systems.
- **Availability** - this is to ensure that systems and data are available to an Organisation when they are required.

The main objective of this project is to assess the baseline security condition of systems, implement hardening, and measure the difference in the level of improvements.

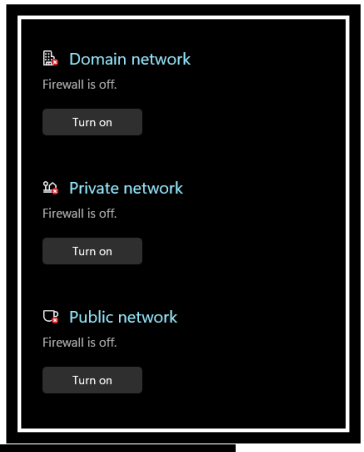
2. Initial Risk Assessment (Before Hardening)

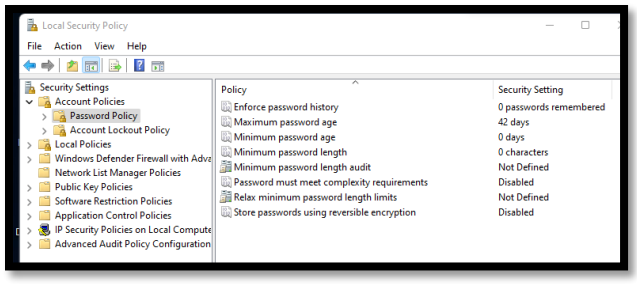
Upon the complete initial assessment, the following security weaknesses were identified:

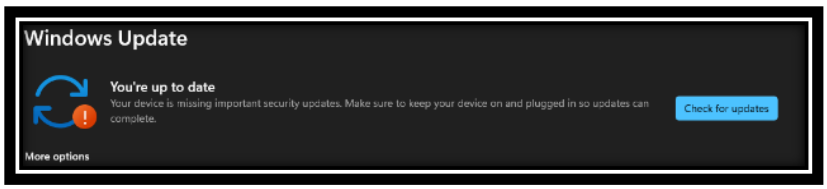
- Weak password policy
- Firewall not completely configured
- BitLocker encryption not enabled
- Unneeded services running
- Pending Windows updates
- No verified backup configuration

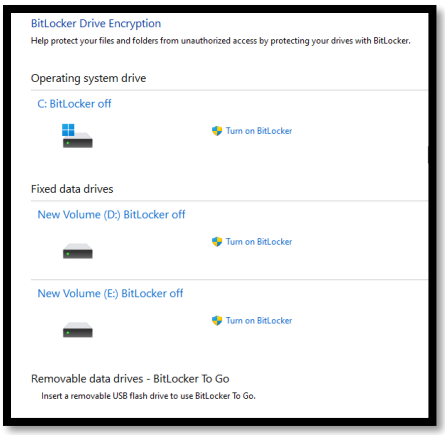
Identified Risks:

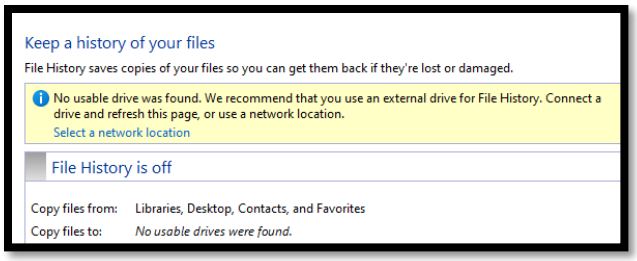
- Unauthorized access to the systems
- Has the potential for malware infection
- Increased potential for theft of customer data
- Utilization of known security vulnerabilities
- Have the potential to create an unstable system and result in data loss.











3. Security Controls Implemented

Control 1: Enabled BitLocker Drive Encryption

CIA Principle: Confidentiality

The system drive has been encrypted with BitLocker Drive Encryption to protect confidential data from unauthorized access by an unauthorized individual who has physically obtained access to the drive.

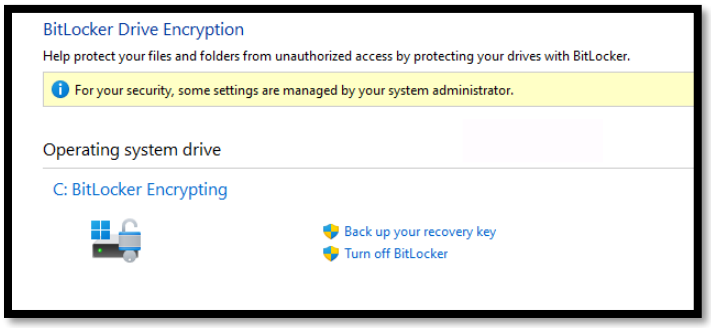


Figure 1: BitLocker Drive Encryption activated to protect system drive data.

Risk Mitigated: Loss of data through theft or unauthorized physical access.

Control 2: Configured Strong Password Policy

CIA Principle: Confidentiality

Using the Local Security Policy (secpol.msc), the following password policy was implemented:

- Minimum Password Length: 12 characters
- Password Complexity Requirements: Enabled
- Password History Enforcement (Last 6 Passwords): Enabled.

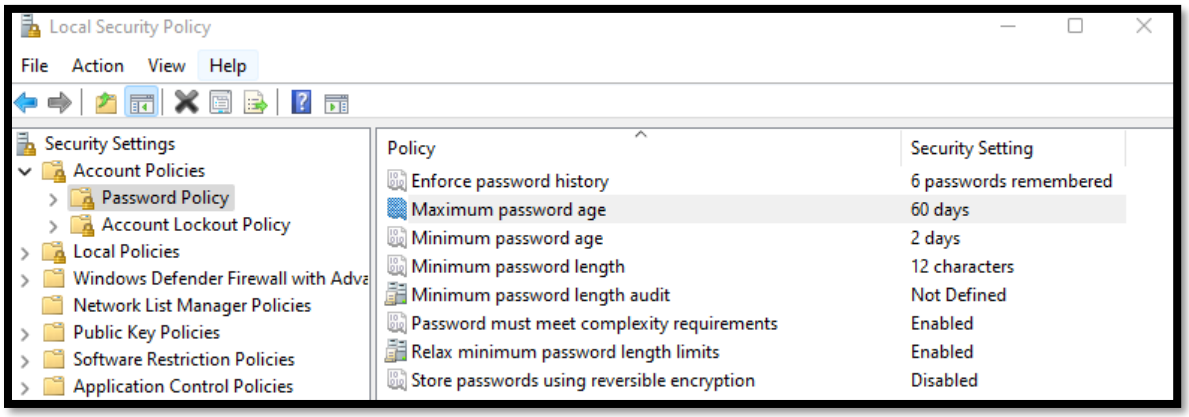


Figure 2: Password Policy configured with a minimum of 12 characters and complexity enabled.

- **Risk Mitigated:** Credential compromise and brute force attacks.

Control 3: Windows Defender Firewall has been Activated and Set Correctly

CIA Principle: Integrity

Windows Defender Firewall has been enabled on both private and public networks.

To protect from unauthorized access and exploitation of a network, I have implemented a custom inbound rule for the Telnet protocol (Port 23).

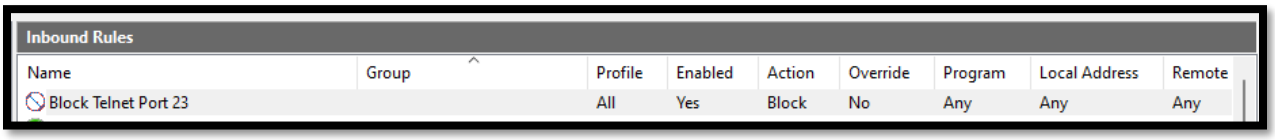


Figure 3: Windows Defender Firewall enabled with custom rule blocking Port 23 (Telnet).

Risk Mitigated: Unauthorized Remote Access and Exploiting of the Network.

Control 4: Windows Defender Protection Has Been Activated

CIA Principle: Integrity

The following Windows Defender protections were enabled:

- Real-Time Protection
- Cloud-delivered Protection
- Tamper Protection.

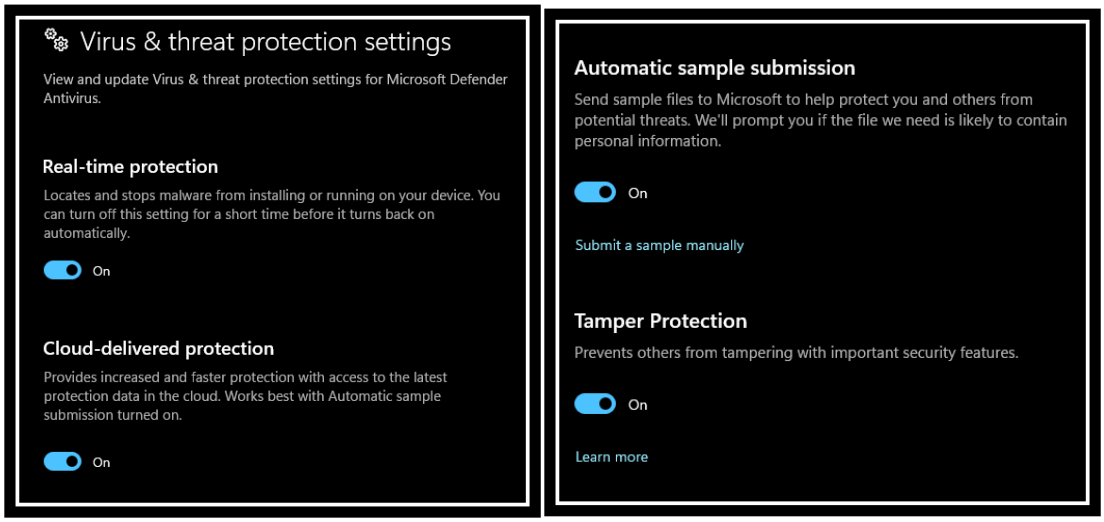


Figure 4: Windows Defender security protections activated.

Risk Mitigated: Change to Existing System Resources and Infection of Malware.

Control 5: Updated Windows Operating System

CIA Principle: Integrity

The complete set of installed updates is current according to Microsoft to patch defects and increase reliability.

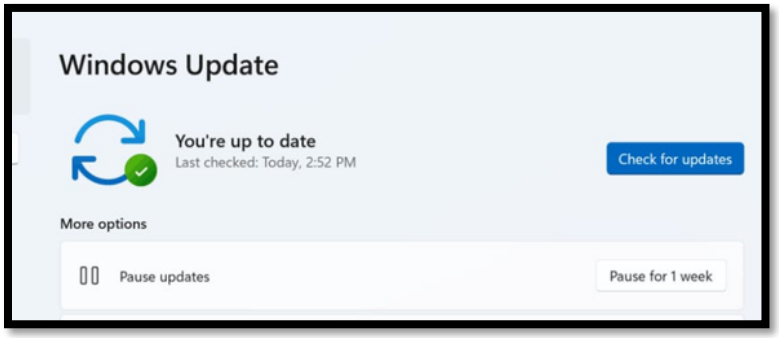


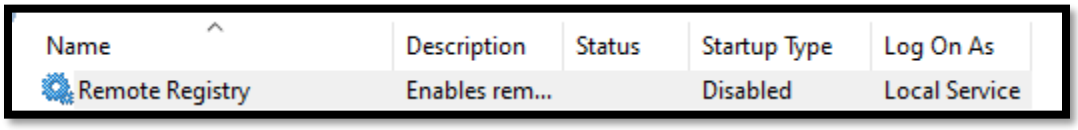
Figure 5: Windows operating system updated with latest security patches.

Risk Mitigated: Exploitation of software defects

Control 6: Disabled Unnecessary Services

CIA Principle: Availability

To help reduce the attack surface area and to increase performance, services that are not used (Remote Registry) have been turned off.



Name	Description	Status	Startup Type	Log On As
Remote Registry	Enables rem...		Disabled	Local Service

Figure 6: Unnecessary service (Remote Registry) disabled.

Risk Mitigated: Lessened exposure to attack and improved system reliability

Control 7: Established Backup Strategy

CIA Principle: Availability

System Restore Points (Create Restore Point) and Backup of Files (Backup Files) have been created to ensure that in the event of a successful Ransomware Attack or system failure, all data can be restored.

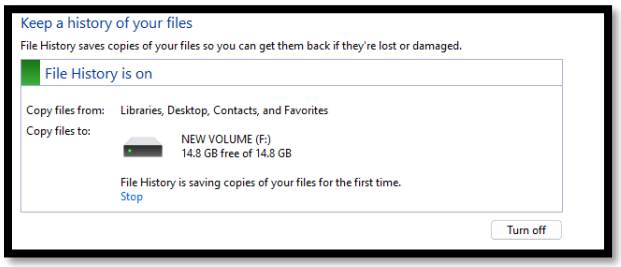


Figure 7: File History backup enabled for automatic file protection.

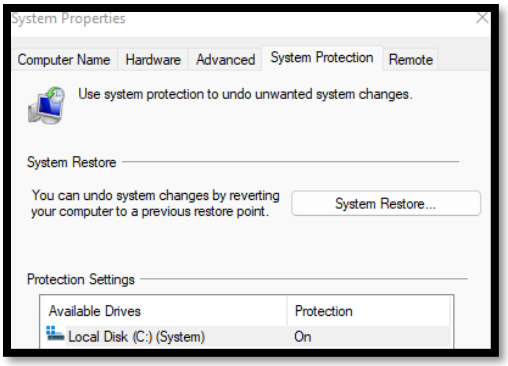


Figure 8: System Protection enabled for Local Disk (C:).

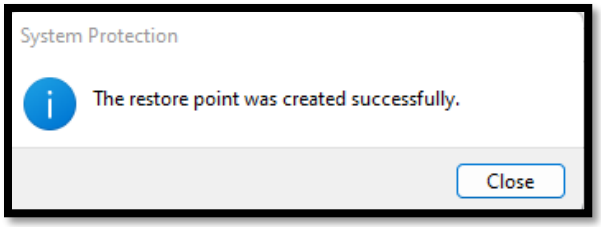


Figure 9: Successful creation of system restore point.

Risk Mitigated: Data loss and extended downtime.